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Valuation of ecological and amenity impacts of an offshore windfarm as a factor in marine planning

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ABSTRACT

Integrated marine planning, which must take into consideration environmental and social impacts, is being introduced widely in Europe, the USA, Australia and elsewhere. Installation of offshore windfarms creates impacts both on local marine ecosystems and the view of the seascape and is one of multiple activities in the marine area that must be addressed by marine planning. The impacts on people's values (and hence welfare) of changes in ecology and amenity that could arise from the installation of a windfarm in the Irish Sea were assessed using a discrete choice experiment administered through an online survey. The ecological changes investigated were: increased species diversity resulting from artificial reef effects, and the effect of electromagnetic fields from subsea cables on marine life; whilst the amenity change was the visibility of offshore turbines from land. Respondents expressed preferences for ecological improvements but had less clear preferences regarding the height and visibility of the turbines. In particular distance decay effects were observed with respondents further away from the coast being less concerned about visual impact created by offshore turbines. Understanding ecological and amenity impacts and how they are valued by people can support the decisions made within marine planning and licensing.

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1. Introduction

In recent years European Union and United Kingdom energy policies for implementation of renewable energy to fulfil carbon emission reduction targets have become one of the main drivers for the development of offshore wind energy. For example, the UK has ambitious plans for the installation of up to 31 GW of offshore wind power (Toke, 2011). Offshore windfarms (OWFs) will mitigate the global environmental effects of climate change through the generation of carbon-free energy, but there are a range of potential local ecological impacts on the marine environment, including the effects of noise, altered sediment distribution and electromagnetic fields, which may disturb or injure marine life (Bailey et al., 2010; Brandt et al., 2011; Fisher and Slater, 2010; Gill et al., 2005; Guillemette and Larsen, 2002; Rothery et al., 2009; Wilson et al., 2010). On the positive side some of the ecological impacts of OWFs can be beneficial. The underwater foundations of turbines (metal or concrete pilings and associated scour protection) have been shown to provide habitat for new species to settle in the OWF site (e.g. Kerckhof et al., 2010). These so-called artificial reef effects increase local biodiversity and, by providing an additional source of food and shelter, can support fish stocks (Reubens et al., 2014) and populations of commercially important brown crab (Hooper and Austen, 2014). However, increases in the abundance or biomass of fish and benthic species are not universal across all sites and species (Ashley et al., 2014), and this introduction of new habitat is not always viewed as positive (Russell et al., 2014). To the best of our knowledge, there have been no attempts so far to assess the level of public concern about and the value of these ecological impacts.

However, the extent to which the sight of the wind turbines on the horizon impairs the view of the seascape has been discussed. The depth limitations on wind turbine installation suggest that in most cases they will be visible from land and thus alter the seascape. It is also possible that substantial amounts of noise could be generated, which might be audible from the shore. These potential amenity impacts have been studied by means of a range of methods including economic valuation and in particular discrete choice experiments (DCE) (e.g. Krueger et al., 2011; Ladenburg and Dubgaard, 2007; Westerberg et al., 2013) and contingent valuation (McCartney, 2006; Müürsepp and Ehrlich, 2012). Most studies find a positive willingness to pay (WTP) of respondents to decrease the

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visual impact of OWFs by moving them further offshore. This indicates that respondents expect a loss of welfare resulting from visible turbines.

Thus, while there are several studies assessing the visual disamenities of offshore turbines there is a lack of studies that value ecological and visual impacts of such installations at the same time. The present study aims to fill this gap by employing a DCE to assess values for local ecological improvements in OWFs and the mitigation of visual disamenities created by the installation of such turbines. Contrary to most previous visual impact studies, which typically use distance from the shore as proxy for visibility, the present study took a different approach by developing the DCE survey around a real future OWF site and varying the potential height of the turbines, whereas the location of the windfarm was already fixed.

The relative importance that the general public attached to each of these effects was studied in a case study in UK waters in the Irish Sea. The assessment of welfare effects resulting from changes in the marine environment can feed into integrated marine planning, which is being widely introduced in Europe through the recent Marine Planning Directive, the USA, Australia and elsewhere. In the UK, marine planning was introduced as a requirement under the Marine and Coastal Access Act (2009) and the Marine Scotland Act (2010) and must take into account a wide range of policies and regulation. The same acts also control licensing consents for offshore infrastructure. Installation of marine energy exploitation devices, including OWFs, is considered as one of multiple activities in the marine area which must be considered within marine planning and for which licensing consents are required. Both marine planning and licensing must take into consideration environmental and social impacts including the wider welfare effects.

When social welfare is affected by the provision of goods or services which are not traded in markets, such as environmental and amenity goods, no market prices can be used for welfare analysis. Therefore, non-market valuation techniques are required to value those ecological and amenity effects of an OWF in support of marine planning and licensing decision-making. The survey-based discrete choice experiment (DCE) approach (e.g. Hanley et al., 1998; Louviere et al., 2000) is one way to asses use and non-use values of non-market environmental and amenity goods. Respondents are required to choose between typically two policy options, each of which will lead to a change in some environmental and/or amenity quality attributes at a cost to the respondent, and a status-quo option which comes at zero cost. In these choice tasks, the policy options are described in terms of a set of choice attributes, which specify the effects of hypothetical management measures. From respondents' stated choices, the value they attach to the different choice attributes can be inferred and expressed as their WTP. WTP can be estimated for each choice attribute. It is an indicator of the change in well-being that respondents expect from a change in the provision of that particular choice attribute.

There are a large number of DCE studies that value the visual disamenities created by an OWF (e.g. Dimitropoulos and Kontoleon, 2009; Krueger et al., 2011; Ladenburg and Dubgaard, 2007; Strazzera et al., 2012; Vecchiato, 2012; Westerberg et al., 2013).[1] In one of the first such studies, Ladenburg and Dubgaard (2007) found significant positive willingness to pay (WTP) of Danish respondents for moving turbines further offshore and reducing noise levels. The set of attributes used in this study did not include any ecological components but focused entirely on technical specifications of the windfarm, such as distance from the shore and number of turbines. Krueger et al. (2011) conducted a similar DCE survey valuing visual disamenity of a windfarm in Delaware, USA. Results showed that while there was significant WTP for moving wind turbines further offshore, the inclusion of certain programmes to fund beach nourishment and development of green energy did not affect choices of respondents living at or close to the coast. It appears that in the presence of visual impacts, respondents were indifferent towards the latter (somewhat more environmental) issues. Westerberg et al. (2013) conducted a choice experiment with tourists in the South of France to investigate whether visual disamenities could be compensated with recreational values created by turbine foundations as artificial reefs (thus providing opportunities for scuba diving, snorkelling and sea-angling). Survey responses indicated that some of the disutility from the visual impact could be compensated by a coherent environmental policy around the windfarm fostering the development of artificial reefs. However, the assessment of this type of ecological impact in this study was only geared towards recreational use of the sea by tourists. Also, respondents to this survey were tourists rather than local residents and interviews were conducted on beaches, which makes selection bias likely. The present DCE study sampled from the whole resident population of a region potentially impacted by OWF development. In a review of some of the above studies Ladenburg and Lutzeyer (2012) found that (1) age, (2) gender, (3) level of education, (4) frequency of visits to the coast, (5) experience with offshore windfarms and (6) location of residence frequently affect the valuation of visual impact in most of the studies. The present study tested the effects of these respondent-specific variables on preference for both amenity and ecological windfarm attributes.

2. Methodology

2.1. Development of valuation scenario and choice attributes; questionnaire design

As a first step in the preparation of the main survey, 60 semi-structured interviews regarding the knowledge of and attitude towards offshore wind energy developments in the UK were conducted with members of the general public in locations near the Irish Sea coast (Liverpool, Cumbria, Lancashire and North Wales). In a second step, four focus group meetings were conducted in Anglesey, Liverpool and Rhyl (Fig. 1). These meetings served to explore which detrimental and beneficial environmental impacts of the installation of OWFs were perceived by people, and to help formulate the valuation scenario and the selection, and description, of the choice attributes. The preliminary questionnaire was then tested in an online pilot survey ($N = 90$) and modified further based on the findings.[2]

In the final valuation scenario, respondents were informed that a windfarm will be installed in the Irish Sea between Anglesey and the Isle of Man (Fig. 1). During the focus groups it had become clear that energy generation, distribution and pricing was a very sensitive topic and so, in order to pre-empt protest responses it was stressed repeatedly that both the decision to build the OWF and the anticipated power output (and thus the supplier's potential profit) were fixed.

The study employed three non-monetary and one monetary choice attributes (Table 1). The first attribute – enhanced biodiversity – was measured in numbers of additional species that are expected to settle in and around the windfarm once it is installed. This attribute conveyed the idea of artificial reef effects

[1] Beyond the literature on visual impact of offshore wind turbines, there is a range of studies on such installations on land (e.g. Alvarez-Farizo and Hanley, 2002; Bergmann et al., 2006, 2008; Meyerhoff et al., 2010), which will not be reviewed here.

[2] Respondents in the pilot survey were drawn from the same panel as the sample for the main survey to ensure that insights gained in the pilot were applicable to the population in the survey area.

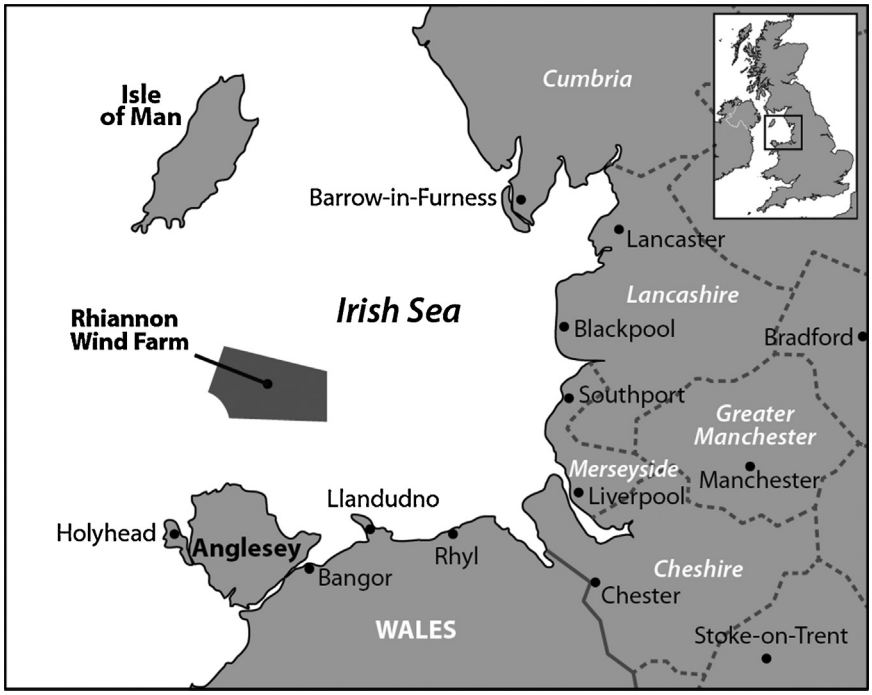

Fig. 1. Future location of the Rhiannon offshore wind farm in the Irish Sea and surrounding urban centres.

Table 1
Choice attributes for discrete choice experiment concerning offshore wind farm (OWF) installation in the Irish Sea (status quo or default design in italics).

Attribute	Description	Levels
Enhanced biodiversity	Number of additional species to settle in and around the new OWF	*No change*; 10 more species; 30 more species
Turbine height and visibility	Height of the wind turbines in metres and coastal strips from which they would be visible	*180 m – visible from Anglesey and the Isle of Man*; 240 m – visible from Anglesey, the Isle of Man and Cumbria; 300 m – visible from Anglesey, the Isle of Man, Cumbria and Liverpool
Electromagnetic fields from cabling	Depending on how deep the cables are buried under the seafloor there is or is not any impact on marine mammals.	*Impact – Cabling buried at 1 m*; No impact – Cabling buried at 2 m
Additional tax	Additional tax to be paid annually by every household to fund alternative windfarm design	*£0*; £5; £10; £25; £50

which have been shown to be caused by the introduction of hard structures into areas with mostly sandy seafloor. The different scenarios altered the form of the turbine foundations, thus resulting in the attraction of different numbers of additional species to the confines of the OWF. The attribute levels chosen were based on a review of monitoring reports that examined the number of species present on OWF foundations (Bouma and Lengkeek, 2012; Bunker, 2004; Kerckhof et al., 2010; Bruijs, 2010). The increase in biodiversity was presented to respondents as a positive change, because the attraction of edible species such as mussels and crab increases the potential of the area to provide enhanced services including food provision and recreation, which are often the drivers for the creation of dedicated artificial reefs (Baine, 2001). Also, the relative footprint of the turbines is small compared to the total area of the OWF, and the conclusions from monitoring reports suggested that the local benthic communities had not been negatively impacted as a result of OWF construction (Lindeboom et al., 2011; Wilson et al., 2010). In contrast to the concerns of conservationists, there was no evidence to suggest that the introduction of additional species was significantly impacting the original habitat, and therefore no reason to present increasing biodiversity as a negative change. In describing the biodiversity attribute, care was taken to avoid naming any particular species as it has been shown that people place a particular value on charismatic and rare species (Jacobsen et al., 2008; Christie et al., 2006), and so the survey design sought to avoid that bias.

The second attribute was non-ecological and related to the height and visual impact of the turbines. Given that the electricity output of the whole windfarm was fixed, it could either have a greater number of relatively small turbines (180 m high) or a smaller number of higher turbines (240 m or 300 m high) (Celtic Array Limited, 2012). A final decision on turbine height had not been made at the time of the survey. Rather than using photo montages of wind turbines of different height, in this study it was specified that windfarms employing turbines of different height would be visible from different sections of the Irish Sea coast (Table 1).[3] The third attribute described the potential impact on marine life of electromagnetic fields (EMF) generated by cables. At current standards, such cables are buried at about 1 m below the sea floor (Wright et al., 2002). At this depth, there are still EMFs that can be felt by certain benthic species (Gill et al., 2005). In order to fully absorb any impact, cables should be buried at least 2 m below the sea floor (Normandeau et al., 2011). It was stressed on the choice card that at this depth there would be no EMF impact, so that respondents effectively had the choice between impact or no impact.

The last attribute was the cost of implementing the alternative wind farm designs, i.e. increasing artificial reef effects or

[3] The use of photo montages by windfarm developers presenting the impact of turbines has been criticised because their design can vary considerably (e.g. MacDonald, 2007).

preventing the impact of EMFs from transmission cables. It was specified as an addition to the existing, compulsory council tax that would be paid by every household in the study region over a period of five years to fund alternative turbine and cabling designs for the Rhiannon OWF. The experimental design was based on information from the online pilot survey. Coefficients indicating the influence of the choice attributes on choices were calculated from responses to the pilot survey applying a mixed logit model (see next subsection). These coefficients were used as priors when developing the choice experiment design with a Bayesian D-efficient design (Scarpa and Rose, 2008) in the software package Ngene (ChoiceMetrics, 2012). The design was restricted to exclude policy options which yield the status quo level for each non-monetary attribute at positive cost, since this option would be dominated by the status quo, which provides no change at zero cost. The final set included 24 choice tasks separated into four different blocks. Respondents were randomly allocated to one block and complete only the six choice tasks in that respective block to keep the cognitive burden at a reasonable level. Each choice task contained two alternative management plans at different cost levels (Options A and B) and a 'no change' option at zero cost (Option C).

The final questionnaire consisted of three sections. Section A contained some questions about the respondent's experience with the Irish Sea and general attitudes towards OWFs. Section B introduced the planned Rhiannon windfarm and the potential ecological and amenity impacts, i.e. the choice attributes. Respondents were instructed to complete six choice tasks and were reminded to take their household budget into account when stating their most preferred option. Subsequent to the choice tasks, this section further included some attitudinal questions about the proposed OWF and the relevant impacts. Section C featured socio-demographic questions in order for the analyst to be able characterise respondents. The questionnaire is available from the authors on request.

The main survey was conducted online between November 2013 and January 2014 via a market research company under close supervision of the researchers. Survey respondents were sampled from an existing panel owned by the market research company. Respondents were chosen according to quota for gender, age group and county of residence to be representative of the resident population of the area adjacent to the Irish Sea.

2.2. Econometric analysis of choice data

The standard approach for the analysis of the choice data is the random utility model (RUM) (McFadden, 1974; Train, 2009). This model relates observed or stated choices to individual utility generated by the attributes of the chosen option. The starting point of the model is the formulation of the utility of respondent n from selecting option i out of a set of options $j = 1, \ldots, i, \ldots, J$ in choice task t as

$$U_{nit} = \beta_n' x_{nit} + \epsilon_{nit}. \tag{1}$$

U_{nit} is the indirect utility function of respondent n. x_{nit} denotes a vector of observable attribute levels of option j and respondent characteristics. β_n is a coefficient vector of these attributes, some or all elements of which are assumed to be random variables, and thus varying across respondents in the mixed logit model (Revelt and Train, 1998). Conversely, in the conditional logit model all the elements are assumed to be constant across respondents, i.e. $\beta_n = \beta \, \forall \, n$. Besides the observable part of utility there is an unobservable component represented by the idiosyncratic error term ε_{nit}. Assuming that respondent n chooses the option which generates the highest utility and that ε_{nit} follows a type I extreme value distribution, the probability P_{nit} that respondent n chooses alternative i over all other alternatives in choice task t is

$$P_{nit} = \frac{\exp(\beta_n' x_{nit})}{\sum_{j=1}^{J} \exp(\beta_n' x_{njt})}. \tag{2}$$

Both conditional and mixed logit models produce estimates of the coefficient vector β_n, the elements of which can be interpreted as the average utility weights for the different choice attributes. These indicate the influence of the different attributes on stated choices. Both model specifications allow for the analysis of explicit preference variation across respondents by interacting respondent-specific variables with choice attribute variables (Train, 2009). Since conditional logit cannot account for unobserved preference heterogeneity, the analysis relied heavily on the mixed logit model. The coefficients of all but the attribute indicating the cost of the choice option to the household were assumed to be normally distributed, so both the mean and standard deviation of coefficients for the respective attributes are reported. It is possible to assume other forms of distribution for the choice attribute coefficients, such as uniform or log-normal, but in this study it was not clear a priori which sign of the coefficients can be expected, so it was decided to apply the most commonly used normal distribution. Since, on the contrary, it could be expected that the cost coefficient will be negative, it was assumed to be fixed across respondents. Respondent-specific variables were included in the models in interactions with a dummy variable indicating whether the status quo or one of the policy options was chosen. The respondent-specific variable 'distance to the coast' was interacted with all choice attribute variables.

The coefficient vector β_n was used to calculate WTP. Since the attribute-specific variables also included a cost component for which the estimated parameter is β_{cost}, mean WTP for any non-monetary choice attribute k was calculated as

$$WTP_k = -\frac{\beta_k}{\beta_{\text{cost}}} \tag{3}$$

where β_k denotes the coefficient of the kth attribute. When the mixed logit model is used, β_k represents the estimated mean of the distribution of the kth attribute. 95%-confidence intervals of mean WTP estimates were calculated based on the bootstrapping approach by Krinsky and Robb (1986). Bootstrapping has to be used because WTP is defined as the ratio of two estimated coefficients as in (3).

3. Results

3.1. Sample characteristics (demographic and attitudinal variables)

Overall 519 questionnaires (one per household) were completed. Respondents in the sample reflected the structure of the resident population in the area surveyed, although there were some limitations (Table 2). In particular, the sample contained fewer respondents from Greater Manchester, but as the study concerned changes in the offshore marine environment, this oversampling of the coastal population did not appear to pose a threat to the validity of results. The choice models considered 19 variables, reflecting the choices made, demographic and spatial variables (Table 3).

3.2. Estimated WTP for OWF characteristics

Table 4 reports both a conditional and a mixed logit model and associated WTP estimates considering the survey sample as a whole. In the conditional logit, the cost coefficient (COST) was negatively affecting choice, whereas the coefficients of different

Table 2
Demographic characteristics of survey respondents as a percentage of the total sampled; Office of National Statistics (ONS) 2011 census data are presented to indicate the survey's regional representativeness.

Variable	*N*	Share/mean	ONS 2011 census[a]
Female	519	57.6%	50.9%
Age (in years)	519		
16–24		14.8%	16.4%[b]
25–44		39.9%	35.6%[b]
45–64		34.9%	35.6%[b]
65 and above		10.4%	12.3%[b]
Household income (after tax)	514	£2396	
Up to ONS level 3 education[c]	501	67.2%	67.2%[d]
County (Place of residence)	519		
Wales[e]		18.5%	15.3%
Cheshire		13.7%	12.3%
Merseyside		16.0%	16.6%
Greater Manchester		17.9%	32.2%
Lancashire		19.1%	17.5%
Cumbria		14.8%	6.9%

[a] 2011 Census data for the survey area only.
[b] Relative to resident population aged 16–75.
[c] Respondents with no qualifications i.e. no GCSE, O-, A-levels, Diploma, NVQ or any other technical or academic qualification.
[d] In England and Wales.
[e] Only includes areas from which respondents were sampled.

increases in species diversity (SPEC10 and SPEC30) and the prevention of the impact of EMF of cables (EMI) affected choices in a positive way. This model detected clear preferences for these attributes, but did not detect any significant effect of turbine height and visibility since the coefficients HEI240 and HEI300 were not significant (Table 4). Estimated WTP for an increase of 10 or 30 species (£12.91 and £15.84, respectively) and the prevention of the impact of EMF on marine life (£26.49) were significantly positive. On average, respondents were indifferent towards the height and visibility of the turbines and were not willing to pay for either lower and less visible or for higher turbines as indicated by the insignificance of the respective coefficients in this model.

The mixed logit model (Table 4), which assumed the coefficients of all non-monetary attributes to be normally distributed, gave similar results to the conditional logit model. WTP for an increase in the number of species by 10 and 30 in the windfarm (£7.25 and £14.83, respectively) and the prevention of electromagnetic impact of the cables (£25.97) were positive. However, this model also found WTP for higher turbines at 240 m to be significantly negative (£-8.23) although the WTP for 300 m high turbines was zero. The fit of this model to the data was far better than in the conditional logit model as indicated by the higher adjusted ρ^2 and the lower BIC stressing the fact that preferences for the attributes are heterogeneous within the sample. This stems from the ability of the mixed logit model to set some attribute coefficients as random. The model did show some unobserved preference heterogeneity for ASC_CHANGE, SPEC30 and EMI as indicated by the significant standard deviation of these coefficients (Table 4), i.e. for these choice attributes, there was significant heterogeneity of preferences across respondents. Regarding turbine height and visibility, however, preferences were found to be homogeneous with standard deviation of the coefficient estimates not significantly different from zero.

3.3. Determinants of choices

Detection of subgroups of respondents that might exhibit differing preference patterns required respondent-specific variables that affect choices to be interacted with a dummy variable indicating the management option (ASC_CHANGE) or with the choice attribute variables. The mixed logit model interacting ASC_CHANGE with demographic variables is presented as Model 1 in Table 5. Attribute coefficients were basically unchanged compared to the mixed logit model in Table 4. The age of the respondent, reported in different age groups, was affecting choices most strongly. Compared to the reference group of 16–24 year olds, all older age groups had a higher likelihood of preferring the default design at no cost (AGE_2, AGE_3 and AGE_4). This preference for the status quo and against the proposed measures intensified with increasing age as the absolute value of the coefficient increased. Apart from age, only membership in an environmental organisation (ENVORG) significantly affected choices. Respondents with such membership were more likely to prefer any of the alternative windfarm designs and their ensuing consequences. Household income (INCOME), level of education (UNI) and gender (MALE) did not affect choices. An additional model, not reported here, found no significant interaction effects between these demographic variables and the turbine height dummies.

The distance between the centroid of a respondent's postcode district and the nearest coastline (DISTANCE) was interacted with all attribute dummies separately (Table 5: Model 2). A significant interaction effect was detected for turbine height and visibility. Models assuming different functional forms of DISTANCE, such as the natural logarithm, were also tested and produced the same negative interaction effect. These different functional forms did not substantially improve model fit, so linear distance was used here. The positive sign of these two interaction coefficients (distance with HEI240 or with HEI300) indicated that the negative effect of turbine height and visibility on choice probability decayed with greater distance from the coast. At the same time, this model produced significantly negative coefficients for both 240 m and 300 m high turbines (i.e. when distance was zero). This means that respondents at the coast derived a significant welfare loss from

Table 3
Description of variables used in the discrete choice experiment regression models with mean values across survey respondents ($N = 519$).

Variable	Description	Mean
ASC_CHANGE	Alternative-specific constant (1 = policy option A or B, 0 = No-change option C)	0.67
SPEC10	Choice attribute increase in species diversity by 10 species (1 = yes, 0 = no)	0.28
SPEC30	Choice attribute increase in species diversity by 30 species (1 = yes, 0 = no)	0.25
HEI240	Turbine height 240 m, turbines visible from Anglesey, the Isle of Man and Cumbria (1 = yes, 0 = no)	0.22
HEI300	Turbine height 300 m, turbines visible from Anglesey, the Isle of Man, Cumbria and Liverpool (1 = yes, 0 = no)	0.24
EMI	Prevention of impact of electromagnetic fields of cables on marine life (1 = Impact prevented, 0 = impact not prevented)	0.40
COST	Cost of the changes in OWF construction as additional tax for the household in £.	14.97
AGE_2	Age of the respondent between 25 and 44 years (1 = yes, 0 = no)	0.40
AGE_3	Age of the respondent between 45 and 64 years (1 = yes, 0 = no)	0.35
AGE_4	Age of the respondent 65 years and above (1 = yes, 0 = no)	0.10
INCOME	Monthly household income of the respondent after tax and benefits in £ GBP	2396
UNI	Respondent has got a university degree (1 = yes, 0 = no)	0.31
MALE	Respondent gender (1 = male, 0 = female)	0.42
ENVORG	Respondent is member of an environmental group (1 = yes, 0 = no)	0.19
DISTANCE	Distance (in km) of respondent *n* to the coast, computed from centroids of postcode districts using GIS	12.01

Table 4
Conditional and mixed logit models (N=497), WTP estimates and confidence intervals of discrete choice experiment.

	Conditional logit				Mixed logit			
	Coef.	Std. Err.	WTP (£)	95%-Conf. Int.	Coef.	Std. Err.	WTP (£)	95%-Conf. Int.
Mean of coefficients								
ASC_CHANGE	−0.529***	(0.108)			0.430	(0.292)		
SPEC10	0.360***	(0.083)	12.91	[6.92 to 18.91]	0.359***	(0.109)	7.25	[2.94 to 11.57]
SPEC30	0.442***	(0.093)	15.84	[9.17 to 22.51]	0.735***	(0.142)	14.83	[9.37 to 20.29]
HEI240	0.009	(0.082)	0.31	[−5.44 to 6.05]	−0.408***	(0.120)	−8.23	[−12.59 to −3.87]
HEI300	0.056	(0.082)	2.02	[−3.82 to 7.87]	−0.162	(0.117)	−3.26	[−7.73 to 1.21]
EMI	0.739***	(0.067)	26.49	[21.50 to 31.48]	1.287***	(0.134)	25.97	[21.03 to 30.92]
COST	−0.028***	(0.002)			−0.050***	(0.003)		
Standard deviation of random coefficients								
ASC_CHANGE					4.914***	(0.364)		
SPEC10					0.058	(0.519)		
SPEC30					0.848***	(0.214)		
HEI240					0.022	(0.328)		
HEI300					−0.291	(0.355)		
EMI					1.585***	(0.152)		
LL_0	−3222				−3222			
Log-likelihood	−3053				−2200			
Observations	8898				8898			
Halton draws	–				1000			
Adjusted ρ^2	0.050				0.313			
BIC	6150				4481			

*** indicate the 1%-level of confidence. Adjusted ρ^2 is computed as $\rho^2 = 1 - (LL_m - k)/LL_0$, where LL_m and LL_0 are the log-likelihoods of the full model, and the intercept-only model, respectively, and k the number of parameters. Bayesian Information Criterion (BIC) is calculated as $BIC = -2LL_m + k \cdot \ln(N)$ with N denoting the number of respondents. The use of BIC is preferred to Akaike Information Criterion because it imposes a stronger penalty on the inclusion of more parameters in the model.

Table 5
The effect of demographic and attitudinal variables on choices of discrete choice experiment (mixed logit models).

	Model 1		Model 2	
	Coefficient	Std. Err.	Coefficient	Std. Err.
Mean of random coefficients				
ASC_CHANGE	1.577*	(0.816)	2.514***	(0.766)
SPEC10	0.374***	(0.112)	0.482***	(0.167)
SPEC30	0.754***	(0.146)	0.727***	(0.216)
HEI240	−0.394***	(0.123)	−0.773***	(0.179)
HEI300	−0.144	(0.120)	−0.583***	(0.179)
EMI	1.311***	(0.135)	1.391***	(0.199)
Standard deviation of random coefficients				
ASC_CHANGE	4.971***	(0.380)	4.801***	(0.379)
SPEC10	−0.001	(0.532)	−0.137	(0.474)
SPEC30	0.879***	(0.215)	−0.991***	(0.229)
HEI240	0.005	(0.343)	0.010	(0.425)
HEI300	0.243	(0.423)	−0.470*	(0.273)
EMI	1.500***	(0.154)	1.722***	(0.175)
Non-random coefficients				
COST	−0.050***	(0.003)	−0.055***	(0.004)
AGE_2[a]	−1.513*	(0.833)	−1.094	(0.792)
AGE_3[a]	−2.395***	(0.870)	−2.166***	(0.813)
AGE_4[a]	−3.185***	(1.119)	−2.799***	(1.070)
INCOME[a]	0.000	(0.000)	−0.0003**	(0.000)
UNI[a]	0.343	(0.578)	−0.058	(0.602)
MALE[a]	0.305	(0.567)	0.080	(0.571)
ENVORG[a]	2.784***	(0.705)	1.116	(0.782)
DISTANCE*SPEC10			−0.001	(0.009)
DISTANCE*SPEC30			0.008	(0.012)
DISTANCE*HEI240			0.018*	(0.009)
DISTANCE*HEI300			0.021**	(0.009)
DISTANCE*EMI			0.003	(0.010)
Log-likelihood	−2056		−1890	
Observations	8448		7728	
Halton draws	1000		1000	
Adjusted ρ^2	0.322		0.316	
BIC	4235		3931	

***, ** and * indicate the 1%-, 5%- and 10%-level of confidence.
[a] Interacted with ASC. Adjusted ρ^2 is computed as $\rho^2 = 1 - (LL_m - k)/LL_0$, where LL_m and LL_0 are the log-likelihoods of the full model, and the intercept-only model, respectively, and k the number of parameters. Bayesian Information Criterion (BIC) is calculated as $BIC = -2LL_m + k \cdot \ln(N)$ with N denoting the number of respondents. The use of BIC is preferred to Akaike Information Criterion because it imposes a stronger penalty on the inclusion of more parameters in the model.

higher and more visible turbines, which was higher than that of respondents living further inland. There was no such spatial effect of the other choice attributes. Preferences for an increase in species diversity in the OWF (SPEC10 and SPEC30) and the prevention of EMF impacts (EMI) did not show any spatial pattern. Somewhat counterintuitively, Model 2 in Table 5 also found a significantly negative income effect on ASC_CHANGE, indicating that respondents with higher incomes were more likely to prefer the default design at no cost and its consequences.

4. Discussion

Results of the discrete choice experiment models showed significantly positive WTP of a sample of respondents from Northwest England and Wales for an increase in species diversity and the prevention of EMF impacting marine life. Yet, the visibility of turbines from different stretches of the coast had only limited influence on choices. While the conditional logit model found respondents to be totally indifferent towards this feature of the OWF, the mixed logit model detected significant WTP to prevent only 240 m high turbines. This latter result is partly in line with previous studies indicating that respondents expect a welfare loss from more visible turbines (e.g. Ladenburg and Dubgaard, 2007; Krueger et al., 2011). Nevertheless, WTP regarding turbines of 300 m in height was again zero, partially supporting the finding of the conditional logit model that respondents' choices were not influenced by this design feature of the OWF. Only after controlling for the effect of distance from the coast could a negative effect on welfare of both levels of higher and more visible turbines be detected. This welfare loss was particularly strong for respondents in coastal locations and decayed with increasing distance from the coast.

Based on these results, the main research question of this study, which was to determine the relative importance of amenity and ecological impacts of an OWF on welfare, can be answered. While the proposed ecological improvements in the OWF significantly affected choices independent of model specification, there was no general effect of turbine height and visibility in the conditional logit and only for 240 m high turbines in the mixed logit. The negative visual impact, found to be substantial in previous studies, did not seem to matter in the same way to respondents in this study where ecological impacts were simultaneously studied. Moreover, positive WTP for the greatest ecological improvements brought about by specific windfarm design by far exceeded negative WTP for accepting more visible turbines in the mixed logit model. Even though the attributes were measured and presented on different scales and units, together with the partial lack of significance of the height coefficients the data indicated a prioritisation of concern by the public of ecological over amenity impacts. Out of the variables that are usually found to influence preferences for reduced OWF visibility (Ladenburg and Lutzeyer, 2012), only age and distance to the coast significantly affected choices. Several other effects of demographic variables on preferences were tested but could not be found in this study.

Although the scenario description clearly stated that the windfarm would produce the same amount of electricity independent of its design, it is possible that some respondents associated higher turbines with higher output. This assumed relationship was mentioned several times during preparatory focus group discussions and might explain the smaller coefficient of HEI300 compared to HEI240. Unfortunately, this hypothesis could not be tested with the present data.

Westerberg et al. (2013) found that certain groups of respondents appreciate the presence of OWF at some distance from the coast. Given the positive interaction effect between turbine height and respondents' distance from the coast, it is possible that respondents living far inland effectively had a preference for higher turbines. However, while the direction of the interaction effect was clear, its extent depends on the functional form of distance (Concu, 2007), so a clear analysis of its form was impossible due to the insufficient size of the dataset. Consequently, the distances from the coast at which the negative coefficients of HEI240 and HEI300 become zero and then potentially positive were not computed.

Regarding the relationship between turbine height and visibility this study assumed a simplistic relationship between higher turbines and greater visibility. The attribute specification did not take into account changing visibility due to light and atmospheric conditions (Bishop and Miller, 2007). Especially with greater distances, these conditions are likely to also decrease visibility. This might serve to explain the average indifference of the sample to 300 m high turbines which would be visible from as far as Liverpool Bay, even though there was an indication of concern over 240 m high turbines in one of the models.

In light of these findings the question arises of which OWF features can and should be valued together in one survey. Evidence for or against the presence of an embedding effect in DCE surveys, the finding that the same good is valued differently alone or in connection with other goods, is mixed (Jacobsen et al., 2011). Further research is needed to investigate the presence of such an effect in discrete choice experiments valuing visual effects of OWFs. The results of this study also call into question the justification of adding up values from different studies especially if used in support of policy development, such as marine planning or management and licensing decisions. If, for instance, WTP for a reduction of visual disamenity of offshore turbines from one survey is compared to WTP for ecological improvements from another survey, the results are likely to be different from the present study. The reason for this is that implicit trade-offs might be at work, which can only be uncovered by simultaneously valuing both effects of the OWF.

5. Conclusions and policy implications

The aim of the present study has been to simultaneously value ecological and amenity effects of a proposed future OWF in UK waters of the Irish Sea. Taken together, the findings are relevant to the future policy and regulatory decisions concerning planning and licensing of OWFs because they demonstrate that non-market welfare effects of such installations are not only derived from the visual impact. Local ecological effects, which, from a welfare perspective, had been largely overlooked in the valuation literature, should be included in cost-benefit analyses of OWF projects. The results show that these ecological changes might affect the welfare of households more strongly and in a spatially invariant way compared to the direct visual effects, which decay with distance from the coast. From a natural science and planning perspective, it has been demonstrated that OWFs have potential to serve as marine protected areas (Ashley et al., 2014) and support specific fisheries, such as for crab and lobster (Hooper and Austen, 2014). Beyond these potential direct benefits of OWFs in terms of conservation and fisheries, the present study demonstrates the existence of additional welfare gains from the installation of such turbines.

Such considerations can particularly inform the integrated marine planning which is now being undertaken in the UK and internationally. This approach is expected to take into consideration all stakeholder perspectives and policies and regulations including the environmental, fisheries, energy and transport sectors. The valuation of all types of market and non-market welfare effects and their integration into decision-making and planning has been a considerable gap to date (Börger et al., 2014). Given the increasing spatial competition between different marine

user groups, co-location of different activities will increasingly be a consideration of marine planning (Christie et al., 2014). Therefore, the assessment of trade-offs between different uses and their values will be a crucial element in the planning process. The findings illustrate the importance of fully considering environmental implications in terms of welfare effects of ecological and amenity impacts in order to secure licensing and planning consents for OWF and other future marine energy developments.

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